Sets Reconstructible with Medial Axis

Adam Białożyt

Faculty of Applied Mathematics, AGH University of Kraków, al. Mickiewicza 30, Kraków, 30-059, Poland .

Contributing authors: bialozyt@agh.edu.pl;

Abstract

The medial axis of a closed set is a well established tool in pattern recognition, cherished for its power of reconstruction of shapes. Yet, a precise description of what the shape means in this context is missing in the literature. This article answers the question which sets precisely are reconstructible from the medial axis information. We introduce the notion of a reconstructible point, being the point in the union of open balls of maximal radius, centred at points of the medial axis. The reconstructible points turn to be precisely the ones lying in the convex hull of the initial set and ones in the half-spaces supported by hyper-planes for which the convex hull and set have different intersection. We do not assume any additional structure of considered sets besides them being closed in $\boldsymbol{n}-$dimensional Euclidean space.

Keywords: Medial Axis, Central Set, Set Reconstruction

1 Introduction

The idea of reducing a data set prior to analysis is ubiquitous in science. A celebrated group of such preprocessing techniques is linked to the so-called medial axis transformation. The medial axis of a closed set X is a collection of points in the ambient space with more than one closest point in X (see Definition 1 below and the example in Fig. 2.1). Although the medial axis and other skeletal structures appear in the works of many mathematicians of the XX century (Erdos, Federer, ... [13, 14]), Harry Blum is typically considered to be their originator. In [7] he argues for their feasibility in the fields of pattern recognition. One of the reasons why the medial axis gained its importance is that it properly encodes the shape of X. The medial axis is a strong topological retract and preserves the homotopy groups of X^c. The applications of the medial axis appear predominately in pattern recognition [21, 22]; however, they are also visible in other fields, such as crowd simulations [24] or robotics [18].

The common knowledge is that the medial axis of X paired with the distance function allows one to reconstruct shapes. However, what does the shape really mean here? And are we sure that we can reconstruct any? These questions appear to be suppressed in the literature. In this article, we will look at them in more detail.

The article is organised as follows. In section 2 we provide the setting and notations of our investigations, the definition of the medial axis, and some similar notions. Next, in section 3, we discuss the reconstructibility of points in $\mathbb{R}^n$ and investigate an example that requires for analysis all the presented theorems. In section 4 we quickly comment on the stability of reconstructible points. Lastly, we briefly discuss the possibility of adaptation of the results to other contexts and problems that might arise during that process.

2 Basic definitions

Throughout the article, we assume that X is a non-empty closed subset of $\mathbb{R}^n$ endowed with the Euclidean distance. By X^c we denote the complement of X. Since we will use the symbol $d(\cdot,\cdot)$ to denote the distance to the set, we will stick to the norm notation $\|a-x\|$ to signify the distance of points a and x. By $[a,x]$ (respectively, by $]a,x]$ and $]a,x[$) we denote the closed segment that joins x and a (resp. the segments $[a,x]\setminus\{a\}$ and $[a,x]\setminus\{a,x\}$).

We will denote by $\mathbb{B}(a,r)$ an open ball of radius $r>0$ which is centred at point $a\in\mathbb{R}^n$; by $\mathbb{S}(a,r)$ we will denote the boundary of $\mathbb{B}(a,r)$, that is, the $(n-1)$-dimensional sphere of the same radius and co-centric with $\mathbb{B}(a,r)$.

Whenever in the paper the continuity (or upper and lower limits) of a family of sets or of a (multi)function is mentioned, it refers to the continuity (or upper and lower limits) in the Kuratowski sense. Let us quickly recall that given a family $\{X_t\}_{t\in\mathbb{R}^k}$ of subsets of $\mathbb{R}^n$, a point $x\in\mathbb{R}^n$ belongs to:

$$\limsup_{t\to t_0} X_t \text{ iff } \exists\mathbb{R}^k\ni t_\nu\to t_0,\ \exists X_{t_\nu}\ni x_\nu\to x;$$
$$\liminf_{t\to t_0} X_t \text{ iff } \forall\mathbb{R}^k\ni t_\nu\to t_0,\ \exists X_{t_\nu}\ni x_\nu\to x.$$

Naturally, whenever two partial limits coincide, we call this limit set the Kuratowski limit. More on the Kuratowski convergence and its relation with the medial axes is found in [5, 23].

2.1 Medial axis and other skeletal structures

Despite the popularity of skeleton-like concepts, there is a bit of confusion as to what we should really call the medial axis. The literature varies when it comes to the details of the definition (or sometimes does not provide one at all). These details give rise to similar, however, not exactly the same objects.

To fix the attention, let us call the medial axis nothing else but the locus of points in $\mathbb{R}^n$ that have more than one closest point in X. We will adopt the following notation. For a non-empty closed set X and a point $a\in\mathbb{R}^n$ we denote by $d(a,X)$ the Euclidean distance from a to X, that is, the infimum of $\|a-x\|$ taken over all points x in X.

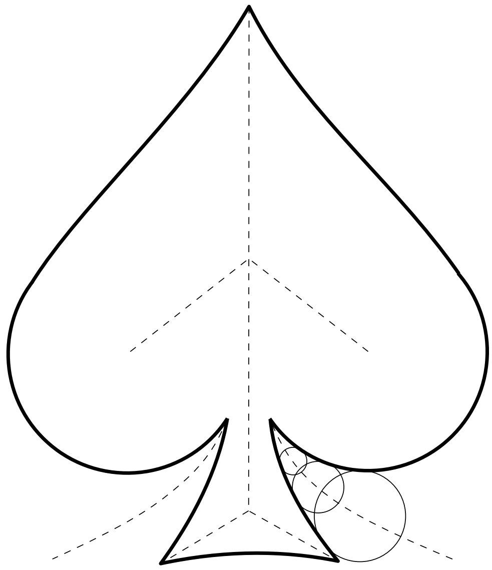

Fig. 1 A medial axis (in a dashed line) of a spade shape. Three maximal balls with centres at points of the medial axis are drawn at the bottom right

$$d(a,X)=d(a):=\inf_{x\in X}\|a-x\|.$$

Next, we collect the points of X closest to $a\in\mathbb{R}^n$ in a set $m(a)$.[1]

$$m_X(a)=m(a):=\{x\in X\mid \|a-x\|=d(a,X)\}.$$

Definition 1 The medial axis of X denoted by M_X is the set of points $a\in\mathbb{R}^n$ for which $m(a)$ has more than one point.

$$M_X:=\{a\in\mathbb{R}^n\mid \#m(a)>1\}.$$

Adopting this point of view has consequences in $M_X\cap X=\emptyset$. The medial axis forms sort of an exoskeleton of a closed set.

2.1.1 Central set

A notion used simultaneously with the medial axis is that of the central set. Its precise construction is as follows. For a non-empty closed set $X\subset\mathbb{R}^n$,

[1] Although $m(a)$ is formally a set of points, we often identify it with a point of $\mathbb{R}^n$ when $\#m(a)=1$

consider a family of open balls contained in the complement of X.

$$\mathcal{B}_X := \{\mathbb{B}(a,r) \mid a \in \mathbb{R}^n,\, r > 0,\, \mathbb{B}(a,r) \subset X^c\}.$$

Naturally, the family $\mathcal{B}_X$ is partially ordered by a relation of inclusion. We call a ball $\mathbb{B}(a,r) \in \mathcal{B}_X$ *maximal* if it is a maximal element of the inclusion relation. That is, for any other ball $\mathbb{B}(a',r') \in \mathcal{B}_X$ we have $\mathbb{B}(a,r) \subset \mathbb{B}(a',r') \Rightarrow a = a', r = r'$.

Definition 2 The central set of X denoted by C_X is the set of centres of maximal balls of the family $\mathcal{B}_X$.

Although the definition seems to construct the same object as Definition 1, in reality the two notions subtly differ. We have $M_X \subset C_X \subset \overline{M_X}$ (see [19]) and either of the inclusions can be strict. In practice, finding the points of C_X is often easier than finding the ones of M_X. Luckily, due to inclusions, if we localise a point $c \in C_X$, we can be sure that $a \in M_X$ can be found arbitrarily close to c. For a more detailed study of the differences between sets C_X and M_X, we refer to [15].

2.1.2 λ-medial axis and scale medial axis

One of the hardships of applying the medial axis transformation is the sensitivity of the latter on the perturbation in the data. Even a small distortion of a set can result in dramatic changes in the medial axis. One of the approaches to tempering the process involves a change in the measurement of the distortion in the medial axis[2] (see [12]). Other treats the distortion of the set as a continuous process[3] (see [5]).

Another well studied approach calls for pruning of the medial axis. We refer here to the so-called $\lambda-$medial axis $M_\lambda(X)$, introduced by Chazal and Lieutier [11]. The $\lambda-$medial axis is the result of removing from M_X the points a with $m(a)$ that fits within some ball $\overline{\mathbb{B}(p,\lambda)}$. The $\lambda-$medial axis is stable under small perturbations of the set X. In general, this approach to stability comes with a price. The $\lambda-$medial axis might not preserve the homotopy type of X^c (so the set reconstructed with $M_\lambda(X)$ might have a different homotopy type than X^c). In particular, Chazal and Lieutier showed that the homotopy type is preserved when the so-called *weak feature size* of X is greater than λ.

2.1.3 Medial axis on Riemannian manifolds

The question of the number of points that realise the distance $d(a)$ also makes sense in the setting of Riemannian manifolds. Changing the surrounding, an interesting phenomenon emerges; one that was never met in $\mathbb{R}^n$. Two points on the manifold might be connected in the shortest way by two different paths. The medial axis of a closed set X on a complete Riemannian manifold is adapted to detect this phenomenon. It collects points that have non-unique shortest paths (geodesics) that connect to X. In this way, it readily generalises the notion of a cut locus of a point.

The study of the medial axis of Riemannian manifolds is drawing increasing attention [1, 17, 8, 25, 20]. However, extreme attention to detail is advised while studying the Riemannian generalisation. The existence of conjugate points or several geodesics with common endpoints complicates the handling of the medial axis.

3 Reconstructible sets

The reconstruction process of $X^c := \mathbb{R}^n \backslash X$ with the medial axis M_X consists of taking a union of open balls centred at points $a \in M_X$ with radii $d(a)$ (see Fig. 3). It is also known by the name of inverse medial axis transform. To make our study of the reconstruction process more structured, we pose the following definition.

Definition 3 Let $X \subset \mathbb{R}^n$ be a closed set. We call a point $p \in X^c$ *reconstructible* if $p \in \bigcup_{a \in M_X} \mathbb{B}(a, d(a))$. We call the set X^c *reconstructible* if every point $p \in X^c$ is reconstructible.

If the context is clear, we will omit the set X used to establish the reconstructibility of a point. In case such precision is necessary, we will signal X by noting its complement that contains the

[2] Rather than measuring the change of M_X and $M_{x'}$ with the Hausdorff distance $d_H(M_X, M_{X'})$, we measure it with its hyperbolic variant.

[3] We treat X' as a Kuratowski limit $\lim X_t$ of a family $\{X_t\}$ with $X_0 = X$. Then the inner-continuity of the medial axes occurs $M_{X'} \subset \liminf M_{X_t}$.

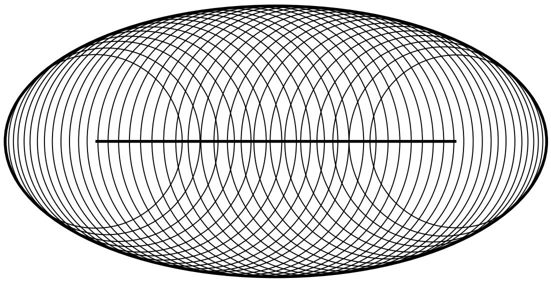

Fig. 2 The reconstruction process of the interior of an ellipse.

point p and write explicitly $p \in X^c$. Let us quickly observe that, without any change in the notion of a reconstructible point, we can use the central set C_X in *lieu* of the medial axis M_X in Definition 3.

Lemma 1 *Let $X \subset \mathbb{R}^n$ be a closed set and let $p \in X^c$. Then there exists $c \in C_X$ such that $p \in \mathbb{B}(c, d(c))$ if and only if p is reconstructible.*

Proof If p is reconstructible, then there exist $a \in M_X$ such that $p \in \mathbb{B}(a, d(a))$. But we have the inclusion $M_X \subset C_X$, so $a \in C_X$.

On the other hand, if there exists $c \in C_X$ such that $p \in \mathbb{B}(c, d(c))$ then $\|p - c\| < d(c)$. The point c can be approximated by a sequence of points $x_n \in M_X$. Since the distance functions (both the point-point and the point-set one) are continuous, for n large enough, we have $\|p - x_n\| < d(x_n)$. For such n we have $x_n \in M_X$ and $p \in \mathbb{B}(x_n, d(x_n))$ so that the point p is reconstructible. □

The main question of this article is which sets are reconstructible with the help of the medial axis. The question might sound almost blasphemous. After all, the medial axis transformation is renown for reconstructing shapes [2, 10, 16, 26]. However, a short glance at the example of a closed unit disc $\mathbb{D} := \overline{\mathbb{B}(0,1)}$ (or, more generally, any convex set) shows us with great clarity that, in fact, not all points of $\mathbb{D}^c$ can be retrieved from $M_{\mathbb{D}}$. Indeed, the medial axis $M_{\mathbb{D}}$ is empty! So, there is no point that can act as a centre of the ball in the union $\bigcup_{a \in M_{\mathbb{D}}} \mathbb{B}(a, d(a))$ and (if we skip the ambiguity of the set theoretic union over an empty family) the reconstruction process fails to bring back any of the points of $\mathbb{D}^c$.

However, the esteem of the medial axis as a shape reconstructor ought to have a source. Let us verify that it can reconstruct at least a part of X^c included in the convex hull of X.

Proposition 2 *Let $X \subset \mathbb{R}^n$ be a closed set and let $p \in X^c$ be a point of the convex hull $\operatorname{conv} X$. Then p is reconstructible.*

Proof Due to the Caratheodory lemma, p is an interior point of at most $n-$dimensional simplex with vertices $v_0, \ldots, v_k \in X$. Assume that $p \notin M_X$ (otherwise, p is surely reconstructible). Thus, we can find a unique closest point of p in X that we denote by x_p.

The idea now is to show that we cannot inflate the ball $\mathbb{B}(p, d(p))$ indefinitely while maintaining its tangency with X only at x_p. Denoting by $p_t := t(p - x_p) + x_p$ we can rephrase that claim more formally as a condition

$$r := \sup\{t > 0 \mid \mathbb{B}(p_t, d(p_t)) \cap X = \emptyset\} < \infty.[4]$$

Observe that $r = \infty$ means in fact that the whole half-space $\{x \in \mathbb{R}^n \mid \langle x - x_p, p - x_p\rangle > 0\}$ is disjoint with X, as it is the union of the family of balls $\mathbb{B}(p_t, d(p_t))$ with $t > 0$. We then have $\langle v_i - x_p, p - x_p\rangle \leq 0$ for $i = 0, \ldots, k$. Recall now that p is the interior point of the simplex spanned by $v_0, \ldots, v_k$. Therefore, we can represent p as the sum $\sum \lambda_i v_i$ with $\sum \lambda_i = 1$, $\lambda_i \in (0, 1)$. However, that leads to a contradiction

$$\begin{aligned} 0 &< \langle p - x_p, p - x_p\rangle = \\ &= \langle \sum_{i=0}^k \lambda_i v_i - \sum_{i=0}^k \lambda_i x_p, p - x_p\rangle = \\ &= \sum_{i=0}^k \lambda_i \langle v_i - x_p, p - x_p\rangle \leq 0. \end{aligned}$$

Finally, as $r < \infty$ the point p_r is well defined and is a centre of a maximal ball in X^c. In other words $p_r \in C_X$. Due to Lemma 1 the point $p \in \mathbb{B}(p_r, d(p_r))$ is reconstructible. □

Remark 1 Proposition 2 fully encapsulates the classical use of the medial axis, i.e. the reconstruction of X^c which are bounded domains. Indeed, when X^c is bounded, we can find an open ball $\mathbb{B}(a, r)$ such that $X^c \subset \mathbb{B}(a, r)$. What follows is $\mathbb{S}(a, r) \subset (X^c)^c = X$ and

$$X^c \cap \operatorname{conv} X \subset X^c \cap \operatorname{conv} \mathbb{S}(a, r) = X^c.$$

[4] Some readers might recognise in the definition of r an idea of the directional reaching radius from [6]. We will use that idea more in the proof of Proposition 3. See also the footnote 6 therein

Therefore, the whole set X^c is reconstructible due to the last proposition.

Having dealt with the points of $\operatorname{conv} X$ we wish to go further; beyond the convex hull. However, to analyse the points of X^c outside the convex hull of X we will still lean on $\operatorname{conv} X$. A context suitable to formalise our investigations is provided by a notion of supporting hyperplane that comes from convex geometry.

Definition 4 Let $X \subset \mathbb{R}^n$ be a set (not necessarily closed). We call a hyperplane $L \subset \mathbb{R}^n$ *a supporting hyperplane of* X if $X \cap L \neq \emptyset$ and X is entirely contained in one of the two closed half-spaces bounded by L. If that is the case, we denote by L^+ the open half-space disjoint with X.

It is a standard fact of convex geometry that any point on the boundary of a closed convex set $X \subset \mathbb{R}^n$ belongs to some supporting hyperplane of X. However, this hyperplane might not be unique. Consider a cone $C = \{(x, y) \in \mathbb{R}^2 \mid y \geq |x|\}$. Then any plane $L_a = \{(x, y) \in \mathbb{R}^2 \mid y = ax\}$ with $|a| \leq 1$ is a supporting plane of C with $0 \in L_a$. If X is not closed, the supporting plane might or might not exist. To witness a convex set without supporting planes, let us take the open unit ball $\mathbb{B}(0, 1)$. An example of a non closed set with supporting hyperplanes at every point of the boundary is given by $\{(x, y) \in \mathbb{R}^2 \mid y < 0\} \cup \{(0, 0)\}$. Moreover, if X is non-empty, closed, and convex, then it is the intersection of the family $\{\mathbb{R}^n \setminus L^+\}$ where L are the supporting hyperplanes of X.

Proposition 3 *Let $X \subset \mathbb{R}^n$ be a closet set and let $p \in X^c$. Let L be a supporting plane of $\operatorname{conv} X$ such that $X \cap L$ is non-convex. If $p \in L^+$ then p is reconstructible.*

Proof Since the intersection $X_L := X \cap L$ is non-convex, we know that M_{X_L} is non-empty (see [19]). Now, to simplify the exposition of the proof, assume, without loss of generality, that $q \in M_{X_L}$, and $L = \{x \in \mathbb{R}^n \mid x_1 = 0\}$. Let us also denote $\eta = (1, 0, \ldots, 0)$; it is a normal vector to L such that $L^+ = \{v \in \mathbb{R}^n \mid \langle x, \eta\rangle > 0\}$. Denote by a, b two distinct points of $m_{X_L}(q)$. We will show that the direction η is in the cone at infinity of M_X.[5]

Let us investigate the points above the segment $]a, b[$ (in the direction η). Take any $\lambda_1, \lambda_2 > 0$ such that $\lambda_1 + \lambda_2 = 1$, $t > 0$, and $x_t \in m(t\eta + \lambda_1 a + \lambda_2 b)$. Note here that since $x_t \notin L^+$ and $\lambda_1 a + \lambda_2 b \in L$ we have $\langle \eta, \lambda_1 a + \lambda_2 b - x_t\rangle \geq 0$. Denote by $v_t := (t\eta + \lambda_1 a + \lambda_2 b - x_t)/\|t\eta + \lambda_1 a + \lambda_2 b - x_t\|$, and define the directional reaching radius[6]

$$r_v(x) := \sup\{\rho \geq 0 \mid \mathbb{B}(\rho v + x, \rho) \cap X = \emptyset\}.$$

We claim that

$$t \leq r_{v_t}(x_t) < \infty.$$

If x_t is not a unique closest point to $t\eta + \lambda_1 a + \lambda_2 b$ in X then the claim holds; $t = \|t\eta\| \leq \|t\eta + \lambda_1 a + \lambda_2 b - x_t\| = r_{v_t}(x_t) < \infty$ since $\langle \eta, \lambda_1 a + \lambda_2 b - x_t\rangle \geq 0$.

If x_t is unique, then the first inequality of the previous paragraph still holds; we need to justify only $r_{v_t}(x_t) < \infty$. Observe again that $r_{v_t}(x_t) = \infty$ means

$$X \subset \{x \in \mathbb{R}^n \mid \langle t\eta + \lambda_1 a + \lambda_2 b - x_t, x - x_t\rangle \leq 0\}.$$

Hence, for $x = a$ and $x = b$ we obtain

$$\langle t\eta + \lambda_1 a + \lambda_2 b - x_t, a - x_t\rangle \leq 0$$

and

$$\langle t\eta + \lambda_1 a + \lambda_2 b - x_t, b - x_t\rangle \leq 0.$$

Multiplying the inequalities by λ_1 and λ_2 respectively and adding them together, we obtain

$$\langle t\eta + \lambda_1 a + \lambda_2 b - x_t, \lambda_1 a + \lambda_2 b - x_t\rangle \leq 0$$

which can be rewritten as

$$\langle t\eta, \lambda_1 a + \lambda_2 b - x_t\rangle + \|\lambda_1 a + \lambda_2 b - x_t\|^2 \leq 0.$$

Since $\lambda_1 a + \lambda_2 b \neq x_t$, we get $\langle t\eta, \lambda_1 a + \lambda_2 b - x_t\rangle < 0$, which is a contradiction.

Hence, the value of $r_{v_t}(x_t)$ is finite, and the point $r_{v_t}(x_t)v_t + x_t$ is well defined. By the standard argument, it belongs to the central set C_X.

Observe now that for any $t > 0$ the point a is closest to $t\eta + a$. Since the multifunction m is upper-semi-continuous, the points x_t converge to a when $\lambda_1 \to 1$. In other words, for all $t > 0$ and $\varepsilon > 0$ there exists a number $\lambda > 0$ such that for any point $x_t \in m(t\eta + \lambda_t a + (1 - \lambda_t)b)$ we have $\|x_t - a\| \leq \varepsilon$.

So choose $\varepsilon > 0$ and $x_t \in m(t\eta + \lambda_t a + (1 - \lambda_t)b)$ and denote $\Lambda_t := \lambda_t a + (1 - \lambda_t)b - x_t$. Then, the normal

[5] A cone at infinity of a closed set $X \subset \mathbb{R}^n$ is given by the directions $v \in \mathbb{R}^n$ such that there exist a sequence of positive numbers α_t and a sequence of points $x_t \in X$ with $\|x_t\| \to \infty$ and $\alpha_t x_t \to v$ when $t \to \infty$).

[6] The directional reaching radius is a concept defined and studied in [6]. It plays the role of a normal curvature for non-smooth sets; together with the reaching radius, defined in the same article and studied later in [3, 4], it characterises the points of the intersection $M_X \cap X$.

vector v_t takes the form $v_t := (t\eta + \Lambda_t)/\|t\eta + \Lambda_t\|$. We claim that

$$\|r_{v_t}(x_t)v_t + x_t\| \to \infty \text{ when } t \to \infty$$

and

$$(r_{v_t}(x_t)v_t + x_t)/\|r_{v_t}(x_t)v_t + x_t\| \to \eta \text{ when } t \to \infty$$

The first convergence follows

$$\|r_{v_t}(x_t)v_t + x_t\| \geq \langle r_{v_t}(x_t)v_t + x_t, \eta\rangle \geq \\ \geq \langle r_{v_t}(x_t)v_t, \eta\rangle - \varepsilon.$$

Indeed, since $\|t\eta + \Lambda_t\| \leq \|t\eta\| + \|\Lambda_t\| \leq t + \Delta$ with Δ that can be picked universally for all t, we have

$$\langle r_{v_t}(x_t)v_t, \eta\rangle \geq t\frac{\langle t\eta, \eta\rangle - \langle x_t, \eta\rangle}{\|t\eta + \Lambda_t\|} \geq \\ \geq \frac{t^2 - t\langle x_t, \eta\rangle}{t + \Delta} \geq \frac{t^2 - t\varepsilon}{t + \Delta} \to \infty$$

The second convergence follows from

$$\left\langle \frac{r_{v_t}(x_t)v_t + x_t}{\|r_{v_t}(x_t)v_t + x_t\|}, \eta \right\rangle \to 1.$$

Let us prove this. Firstly:

$$\left\langle \frac{r_{v_t}(x_t)v_t + x_t}{\|r_{v_t}(x_t)v_t + x_t\|}, \eta \right\rangle = \\ = \frac{\langle r_{v_t}(x_t)v_t, \eta\rangle}{\|r_{v_t}(x_t)v_t + x_t\|} + \frac{\langle x_t, \eta\rangle}{\|r_{v_t}(x_t)v_t + x_t\|}$$

The second term converges to zero, while the first one is equal

$$\frac{\langle r_{v_t}(x_t)v_t, \eta\rangle}{\|r_{v_t}(x_t)v_t + x_t\|} = \frac{r_{v_t}(x_t)(t - \langle x_t, \eta\rangle)}{\|t\eta + \Lambda_t\| \cdot \|r_{v_t}(x_t)v_t + x_t\|} = \\ \frac{1}{\|\eta + \Lambda_t/t\| \cdot \|v_t + \frac{x_t}{r_{v_t}(x_t)}\|} - \frac{\langle x_t, \eta\rangle}{\|t\eta + \Lambda_t\| \cdot \|v_t + \frac{x_t}{r_{v_t}(x_t)}\|}$$

Since $\|\eta + \Lambda_t/t\| \to \|\eta\| = 1$ and $\|v_t + \frac{x_t}{r_{v_t}(x_t)}\| \to \|v_t\| = 1$ when $t \to \infty$, we obtain the sought limit. Thus, we have found points $c_t := r_{v_t}(x_t)v_t + x_t \in C_X$ such that $c_t/\|c_t\| \to \eta$ when $t \to \infty$ and proved that the direction η is in the cone at infinity of M_X.

That is enough to end the proof. For any p in L^+ and a sequence of points $c_n \in C_X$ with $\|c_n\| \to \infty$, $c_n/\|c_n\| \to \eta$ we will have $d(c_n) > \|p - c_n\|$ for n large enough. For any such n the point p belongs to $\mathbb{B}(c_n, d(c_n))$ so it is reconstructible. $\square$

Proposition 4 *Let $X \subset \mathbb{R}^n$ be a closed set and let $p \in X^c$. Let L be a supporting plane of $\overline{\operatorname{conv} X}$ such that $X \cap L \neq \overline{\operatorname{conv} X} \cap L$. If $p \in L^+$ then p is reconstructible.*

Proof If $X \cap L$ is non convex, then the previous Proposition proves that p is reconstructible. Assume without loss of generality that $L = \{x \in \mathbb{R}^n \mid x_1 = 0\}$ and that $X \cap L$ is convex. We will basically modify the proof of the previous Proposition to adopt it to new assumptions. Denote as before $\eta = (1, 0, \ldots, 0)$. Choose a point $w \in (\overline{\operatorname{conv} X} \cap L)\backslash(X \cap L)$. Select $t > 0$ and $x_t \in m(t\eta + w)$. Note that since $x_t \notin L^+$ and $w \in L$ we have $\langle \eta, w - x_t\rangle \geq 0$. Define the directional reaching radius:

$$r_v(x) := \sup\{\rho \geq 0 \mid \mathbb{B}(\rho v + x, \rho) \cap X = \emptyset\}.$$

We claim that

$$t \leq r_{v_t}(x_t) < \infty.$$

If x_t is not a unique closest point to $t\eta + w$ in X then the claim holds:

$$t = \|t\eta\| \leq \|t\eta + w - x_t\| = r_{v_t}(x_t) < \infty$$

since $\langle \eta, w - x_t\rangle \geq 0$.

If x_t is unique, then the first inequality of the previous paragraph still holds and we need to justify only $r_{v_t}(x_t) < \infty$. Observe again that $r_{v_t}(x_t) = \infty$ means

$$X \subset \{x \in \mathbb{R}^n \mid \langle t\eta + w - x_t, x - x_t\rangle \leq 0\}.$$

Here the differences between the proofs start. We do not necessarily have $a, b \in L$ such that w is their convex combination. However, since $w \in \overline{\operatorname{conv} X}$, we can find sequences of points $a_{i,\nu} \in X$ and scalars $\lambda_{i,\nu} \in (0, 1)$ (where $i = 1, \ldots, k$) such that $\sum_i \lambda_{i,\nu} = 1$ and $w = \lim_{\nu\to\infty} \sum \lambda_{i,\nu} a_{i,\nu}$. We follow in the footsteps of the previous proof with $x = a_{i,\nu}$. For each index i we obtain inequalities

$$\langle t\eta + w - x_t, a_{i,\nu} - x_t\rangle \leq 0$$

Multiplying these inequalities by $\lambda_{i,\nu}$ respectively and adding them together, we obtain

$$\langle t\eta + w - x_t, \sum_i \lambda_{i,\nu} a_{i,\nu} - x_t\rangle \leq 0$$

which can be rewritten as

$$\langle t\eta, \sum_i \lambda_{i,\nu} a_{i\nu} - x_t\rangle + \langle w - x_t, \sum_i \lambda_{i,\nu} a_{i\nu} - x_t\rangle \leq 0.$$

Since $\sum \lambda_{i,\nu} a_{i,\nu} \to w$ when $\nu \to \infty$, and $w \neq x_t$ the second term tends to $\|w - x_t\|^2 > 0$. Consequently, we get as a limit $\langle t\eta, w - x_t\rangle < 0$, which is a contradiction.

The proof of the previous proposition approximates now points above $\xi \in X \cap L$. However, we might not find any such ξ under current assumptions. We need a new procedure.

Surprisingly, we can simply go upward from any point of $p \in X$ located in $(\overline{\operatorname{conv} X}\backslash X) \cap L$. Denote $\delta := \min\{d(x, L) \mid x \in m(p)\}$, and call the point above p to be $w_t = t\eta + p$. For the closest point $x_t \in m(w_t)$ the coordinates of $x_t = (x_t^1, \ldots, x_t^n)$ must satisfy

$$|x_t^1| < \sqrt{t^2 + 2\delta t + d(p)^2} - t$$

and

$$\|(x_t^2, \ldots, x_t^n)\| \leq \sqrt{2t\delta + d(p)^2}.$$

We also find that the vector n_t given by

$$n_t := w_t - x_t = ((t+\delta) - x_t^1, p^2 - x_t^2, \dots, p^n - x_t^n)$$

is normal to X at x_t, and points at w_t. For $v_t = n_t/\|n_t\|$ the normalised expression

$$(r_{v_t}(x_t)v_t - x_t)/\|r_{v_t}(x_t)v_t - x_t\|$$

will get steeper and steeper.

Indeed, $r_{v_t}(x_t) \geq t \to \infty$ as we have seen a moment ago. Now, we see that the fraction

$$\frac{r_{v_t}(x_t)v_t + x_t}{\|r_{v_t}(x_t)v_t + x_t\|} = \frac{r_{v_t}(x_t)v_t}{\|r_{v_t}(x_t)v_t + x_t\|} + \frac{x_t}{\|r_{v_t}(x_t)v_t + x_t\|}$$

with the second term of the expression converging to zero when $t \to \infty$. The directional component v_t of the first term is just a scaled vector n_t, and we can clearly see that $n_t/t \to \eta$. So, the whole fraction $\frac{r_{v_t}(x_t)v_t + x_t}{\|r_{v_t}(x_t)v_t + x_t\|}$, being a vector on the unit sphere, must converge to η. Again, η is a direction in the cone of M_X at infinity.

By the same argument as at the end of the previous proof, the point p belongs to some $\mathbb{B}(c, d(c))$ with $c \in C_X$, hence it is reconstructible. □

Corollary 5 *Let $X \subset \mathbb{R}^n$ be a closed set and let $p \in X^c$ be a point of the closure of the convex hull* conv X. *Then p is reconstructible.*

Proof Since $p \in \overline{\operatorname{conv} X}$ it is either a point of conv X and Proposition 2 says that it is reconstructible, or it is a point of $\overline{\operatorname{conv} X} \setminus X$. In the latter case, any supporting hyper-plane L of $\overline{\operatorname{conv} X}$ with $p \in L$ will raise a sequence of balls as in the proof of Proposition 4 with $p \in \mathbb{B}(c_t, d(c_t))$. □

The series of previous propositions exhaust all the possible cases. We have the following characterisation.

Theorem 6 *Let $X \subset \mathbb{R}^n$ be a closed set. Let $\mathcal{L}$ be a family of supporting hyperplanes of $\overline{\operatorname{conv} X}$ such that $X \cap L \neq \overline{\operatorname{conv} X} \cap L$. Then X^c is reconstructible if and only if it is a subset of a union $\overline{\operatorname{conv} X} \cup \bigcup_{L \in \mathcal{L}} L^+$*

Proof By previous Propositions, if X^c is a subset of $\overline{\operatorname{conv} X} \cup \bigcup_{L \in \mathcal{L}} L^+$ then all points in X^c are reconstructible. It is so since $\overline{\operatorname{conv} X} \cap L$ is convex. So whenever $X \cap L$ is not convex, it cannot be equal to $\overline{\operatorname{conv} X} \cap L$. Let us briefly explain why no other point can be reconstructible.

By definition, every reconstructible point $p \notin \operatorname{conv} X$ must be a point of some ball $\mathbb{B}(a, d(a))$ with $a \in M_X$. Pick a point $q \in m(a)$. Now, since the ball $\mathbb{B}(a, d(a))$ is convex, we can connect the points p and q with a segment $]p, q[\subset \mathbb{B}(a, d(a))$. The intersection of this segment with the closure of conv X forms a subinterval $[a, b[\subset]p, q[$. Now, we clearly see that for any supporting plane L of $\overline{\operatorname{conv} X}$ with $a \in L$, the point a is in $(L \cap \overline{\operatorname{conv} X}) \setminus (X \cap L)$. When we pick L to be a hyperplane that separates p and $\overline{\operatorname{conv} X}$ we have the Theorem.

□

Example 1 Let us investigate the reconstructible points of the set X in Fig. 3.

$$X = \{(x, y) \in \mathbb{R}^2 \mid y = \ln x\} \cup \{(-1, 0)\}.$$

First, compute the convex hull of X. It is a subgraph of $y = \ln x$ together with the subgraph of the affine function tangent to $y = \ln x$ and passing through $(-1, 0)$. The latter touches the graph at point $(\xi, \ln \xi)$ with $\xi = e^{W(1)+1}$ where W denotes the Lambert W function. Therefore, the convex hull conv X is is equal $\{y \leq \ln x\} \cup \{y \leq \xi^{-1}(x+1), x \in (-1, \xi]\} \cup \{(-1, 0)\}$. The supporting lines of $\overline{\operatorname{conv} X}$ are $x = -1$, $y = \xi^{-1}(x+1)$, and every line tangent to the graph $y = \ln x$ at points (x_0, y_0) with $x_0 > \xi$. Only the first two have nonempty intersection with $\overline{\operatorname{conv} X} \setminus X$. So, the subset of X^c that is reconstructible by M_X is

$$\left(\{y < \ln x\} \cup \{x < \xi\} \cup \{y > \xi^{-1}(x+1)\}\right) \setminus X.$$

The reconstruction process does not provide information on the shape of X in the region

$$\Omega = \{\ln x < y < e^{-W(1)-1}(x+1),\ x > e^{W(1)+1}\}.$$

As long as $X \cap \Omega = (\operatorname{conv} X) \cap \Omega$ any deformation of X will not affect M_X or the distance function d restricted to M_X. It agrees with old results of Calabi and Hartnett [9] who proved that sets with the same convex deficiency[7] have the same medial axes.

Example 2 (The reconstruction of a smooth curve.) Let X be the set from the Example 1. The medial axis of X consists of two connected components γ_1, γ_2, which are C^1-smooth curves. Denote by $\gamma_i(t)$ their length parameterisations and let $T_i(t), N_i(t)$ denote, respectively, a unit tangent and a unit normal vector to M_X at $\gamma_i(t)$. Then, considering a formula

$$x_i^{\pm}(t) := \gamma_i(t) - d((\gamma_i(t))d'((\gamma_i(t))T_i'(t) \pm \\ \pm d(\gamma_i(t))\sqrt{1 - d'(\gamma(t))^2} N_i(t)$$

[7] Convex deficiency of a closed set Y is $\operatorname{conv} Y \setminus Y$

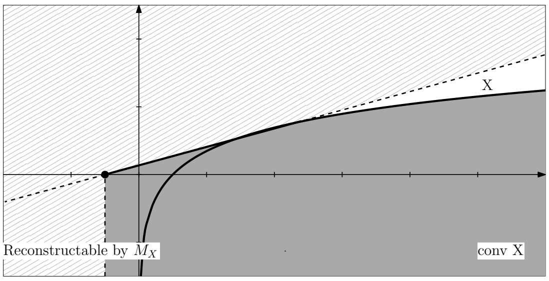

Fig. 3 Example 1 and 2. The shaded region is the convex hull of X. During the reconstruction process with the medial axis, we get back points in $\overline{\mathrm{conv}\, X}$ together with ones in the striped region. The plain white region is not reached by balls centred at points of the medial axis. If we try to recreate the graph of the logarithm function by mapping the points of the medial axis through the multifunction m, we need to be careful. While traversing the connected component of M_X that lies on the upper left we recreate only a part of the graph. The other connected component gives us the full graph of the logarithm.

we get a parametrisation of a part of X. (This reconstruction process of a curve is taken from Giblin [16].) From the analysis in Example 1, we see that the whole curve $y = \ln x$ can be reconstructed fully only when choosing the proper connected component γ_i.

4 Stability of the reconstructible points

The question of stability of the medial axis is prevalent in the literature. How does it apply to the notion of reconstructibility? That is, how does the reconstructible part of X^c behave when we change X slightly? In general, the change can be drastic. A straight line $L = \{y = 0\}$ on the Euclidean plane has an empty medial axis, so the reconstructible part of L^c is empty. If we puncture the line with an arbitrarily small gap $I_\varepsilon :=](-\varepsilon, 0), (\varepsilon, 0)[$, then the entire $(L \backslash I_\varepsilon)^c$ becomes reconstructible.

Luckily, severe changes can appear only outside of conv X. Indeed, Proposition 2 asserts that $X^c \cap \mathrm{conv}\, X$ and $Y^c \cap \mathrm{conv}\, Y$ are reconstructible. If X and Y are close to each other (let us say in the Hausdorff sense), then their convex hulls are close as well. We have a kind of partial stability of the reconstructible points. The only difference in reconstructibility in the region of convex hulls of X and Y comes directly from the symmetric difference of X and Y. However, remember that the complements X^c and Y^c can still vary a lot even for close X and Y,[8]

Next, we can use the Kuratowski convergence theorem to establish another result. Recall that for a family of sets $\{X_t\}_{t \in T}$ such that $X_t \to X_0$ in the Kuratowski sense when $t \to 0$ the medial axes satisfy an inclusion $M_{X_0} \subset \liminf_{t \to 0} M_{X_t}$ (see [5]). We have the following.

Proposition 7 *Let $\{X_t\}_{t \in T}$ be a family of closed subsets of $\mathbb{R}^n$ such that $\lim_{t \to 0} X_t = X$ in the Kuratowski sense and let $p \in X^c$ be reconstrucible. Then there exists a neighbourhood T_0 of $0 \in T$ such that for any $t \in T_0$ the point $p \in X_t^c$ is reconstructible.*

Proof By the definition of a reconstructible point, we can find $q \in M_X$ such that $p \in \mathbb{B}(q, d(q))$. Since $M_X \subset \liminf_{t \to 0} M_{X_t}$, we know that setting a neighbourhood U of q we can find T_0 such that for $t \in T_0$ every M_{X_t} has a point in $q_t \in U$. Set U to be a ball B centred on q and of radius $(d(q) - \|p - q\|)/2$. By shrinking T_0 we can assume $|d_t(q_t) - d(q)| < (d(q) - \|p - q\|)/2$ as well. Then

$$\|p - q_t\| \leq \|q - q_t\| + \|p - q\| \leq$$
$$\leq d(q) - (d(q) - \|p - q\|)/2 \leq d_t(q_t).$$

So, the point p is reconstructible for X_t with $t \in T_0$. $\square$

5 Other skeletal structures and settings

5.1 $\lambda-$medial axis

What subset of X^c can we reconstruct if we restrict the centres of balls in the union $\bigcup_{p \in M_X} \mathbb{B}(p, d(p))$ to the points $p \in M_\lambda(X)$ (see sec. 2.1.2)? In analogy to Definition 3, we propose the following notion of $\lambda-$reconstructible points.

Definition 5 Let $X \subset \mathbb{R}^n$ be a closed set and let $\lambda \geq 0$. A point $p \in X^c$ is said to be $\lambda-$ *reconstructible* if $p \in \bigcup_{a \in M_\lambda(X)} \mathbb{B}(a, d(a))$.

Surely, the $\lambda-$reconstructible part of X^c will be a subset of the part reconstructible in the

[8] Take, for example, $X_k = \mathbb{R} \backslash ((-1, 1) \cup (k, k+1))$ and $Y = \mathbb{R} \backslash ((-1, 1))$. Even though X_k and Y each are contained in an $1-$offset of the other, their complements can be arbitrarily far from each other.

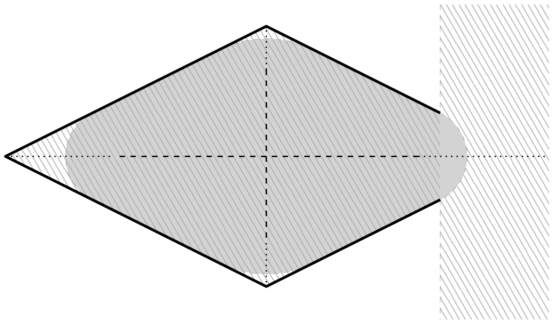

Fig. 4 In general, the set of $\lambda-$reconstructible points forms a proper subset of reconstructible points. In example above, the dotted and dashed lines represent respectively the medial axis and the $\lambda-$medial axis of the shape. Only the grey area is $\lambda-$reconstructible, when λ is set to be slightly larger than the gap on the right side of the shape. It forms a proper subset of the set of reconstructible points, depicted as the striped area.

sense of definition 3. Intuitively, obstructions to $\lambda-$reconstructibility will arise only in the bottlenecks of X that can take the form of tunnels, sharp corners, or gates at the boundary of $\operatorname{conv} X$. Unfortunately, the proofs of Propositions 2 – 4 do not discuss the diameter of $m(a)$. Actually, it is not immediately obvious how to bound the diameter of a ball encasing $m(a)$ in terms of $d(a)$ nor how this diameter behaves for x close to a (we only know that it is upper semi-continuous). It appears that the proofs of the theorems reformulated for the $\lambda-$medial axis need to be a bit more elaborate.

Let $dist(A, B) := \inf\{\|a - b\| \mid a \in A, b \in B$. So far we have only the following weak version of Proposition 4.

Proposition 8 *Let $X \subset \mathbb{R}^n$ be a closed set and let L be a supporting plane of $\operatorname{conv} X$ and denote $X_L := X \cap L$. Choose $\lambda > 0$ and assume that there exists a point $q \in L$ such that $m_{X_L}(q)$ has at least two connected components and that for any two distinct connected components A, B we have $dist(A, B) > 2\lambda$. Then every point $p \in L^+$ is $\lambda-$reconstructible.*

Proof As before, to simplify the exposition of the proof, assume, without loss of generality, that $q = 0 \in M_\lambda(X_L)$, $L = \{x \in \mathbb{R}^n \mid x_1 = 0\}$, and that $d(q, X \cap L) = 1$. Let us also denote $\eta = (1, 0, \ldots, 0)$; We will show that for any $T > 0$ we can find points of $M_\lambda(X)$ in an unbounded cylinder $W_T = \{x \in \mathbb{R}^n \mid \langle x, \eta\rangle > T, \|x - \langle x, \eta\rangle\eta\| < 1\}$.

Denote

$$X_\delta := X \cap \{x \in \mathbb{R}^n \mid d(x, L) \leq \delta\}.$$

We obviously have $X_L = X_0$. Moreover, since X is closed, there is also $\lim_{\delta\to 0} X_\delta = X_0$.

Since $dist(A, B) > 2\lambda$ for any two distinct connected components A, B of $m_{X_L}(q)$, there exists a collection of open sets U_ι, such that $m_{X_L}(q) \subset \bigcup U_\iota$ and

$$\overline{U_i} \cap U_j = U_i \cap \overline{U_j} = \emptyset, \text{ when } i \neq j$$

and points

$$x_\iota \in U_\iota \cap m_{X_L}(q).$$

Actually, since the multifunction m has compact values, we can assume that sets U_ι are still relatively far from each other, meaning $dist(\overline{U_i}, \overline{U_j}) > 2\lambda$ for $i \neq j$.

Since X is closed, we can find δ such that $X_\delta \cap \overline{W_{-1}}$ is contained in the disjoint union $\bigcup U_\iota$. Then by choosing t large enough we can ensure that the intervals $[x_i + t\eta, t\eta]$ are mapped by the multifunction m onto $X_\delta \cap \overline{W_{-1}} \subset \bigcup U_\iota$.

Now, traversing the segments $[x_i + t\eta, t\eta]$ and $[t\eta, x_j + t\eta]$ with $t > 0$ large enough and $j \neq i$, we encounter a discontinuity of the multifunction m. Arguments for which discontinuities occur are points of the medial axis M_X. Moreover, since

$$m(x_i + t\eta) = x_i \in U_i \text{ and } m(x_j + t\eta) = x_j \in U_j,$$

we can say that for (at least) one of the discontinuity points, say ξ, the index ι for which $m(x) \cap U_\iota \neq \emptyset$ needs to change when x is passing through ξ. Since $\lim_{x\to\xi} m(x) \subset m(\xi)$, the set $m(\xi)$ contains points from at least two distinct U_ι, U_κ. The minimal radius of a ball that surrounds $m(\xi)$ will be at least equal to $dist(\overline{U_\iota}, \overline{U_\kappa})/2 > \lambda$. □

When it comes to $\lambda-$reconstructibility of the interior points of X, we conjecture.

Conjecture 9 *Let $X \subset \mathbb{R}^n$ be a closed set and let $p \in \operatorname{conv} X$. Choose $\lambda > 0$ and assume that $d(p) > \lambda$. Then p is $\lambda-$reconstrucitble.*

5.2 Riemannian manifolds

Do we observe the same effects on the Riemannian manifolds? The change in space in which we situate the set X has consequences on the medial axis. First, the medial axis does not contain any more only the points with the non-unique closest point. It also collects the points with the non-unique shortest path to X (consider a single point on a unit sphere to see the difference). Second, the proofs that involve operations on the scalar product must be transported to the tangent space, or geodesics. Last, one should also question the behaviour (and the definition) of supporting hyper-planes.

However, a systematic study of the reconstruction process seems promising. Observe that one of the consequences of Theorem 6 on a compact manifold $\mathcal{M}$ would be that the whole complement of a closed subset $X \subset \mathcal{M}$ is reconstructible.

6 Conclusion

The ability of shape reconstruction is one of the revered traits of the medial axes. However, as we demonstrated at the beginning of the article, not all shapes are fully reconstructed. Therefore, we have defined and investigated the reconstructible points of the set complement X^c. The study shows that the reconstructible points of X^c are precisely those of the convex hull of X and of the half-spaces L^+, where L is a supporting hyperplane of $\overline{X}$ and $X \cap L \neq \overline{\operatorname{conv} X} \cap L$. Some variants of the stability of the reconstructible points.

We proposed the analogue of the reconstructible points for the $\lambda-$medial axis. However, the proofs of our theorems do not adapt easily to the $\lambda-$medial axis. For the $\lambda-$ medial axis, only a weak version of Proposition 3 was proved. We also propose a conjecture about the $\lambda-$reconstructible points in the convex hull of X. Lastly, we hint at the feasibility of the topic on Riemannian manifolds.

Declarations

Funding

This work was partially supported by the Faculty of Applied Mathematics AGH University statutory tasks under the subvention of the Ministry of Science and Higher Education (Poland)

References

[1] P. Albano. "On the cut locus of closed sets". In: *Nonlinear Anal.-Theor.* 125.C (2015), pp. 398–405. DOI: 10.1016/j.na.2015.06.003.

[2] D. Attali, J.-D. Boissonnat, and H. Edelsbrunner. "Stability and Computation of Medial Axes - a State-of-the-Art Report". In: *Mathematical Foundations of Scientific Visualization, Computer Graphics, and Massive Data Exploration.* Ed. by T. Möller, B. Hamann, and R. D. Russell. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 109–125. DOI: 10.1007/b106657_6.

[3] A. Białożyt. "The tangent cone, the dimension and the frontier of the medial axis". In: *Nonlinear Differential Equations Appl. NoDEA* 30 (2020), pp. 1–29. DOI: 10.1007/s00030-022-00833-9.

[4] A. Białożyt, D. Bysiewicz, and M. P. Denkowski. *Directional curvature and medial axis.* 2026. arXiv: 2604.26490 [math.MG].

[5] A. Białożyt, A. Denkowska, and M. Denkowski. "The Kuratowski convergence of medial axis and conflict sets". In: *Ann. Sc. Norm. Super. Pisa Cl. Sci* (2024), p. 21. DOI: 10.2422/2036-2145.202310_002.

[6] L. Birbrair and M. Denkowski. "Medial axis and Singularieties". In: *J. Geom. Anal.* 27.3 (2017), pp. 2339–2380. DOI: 10.1007/s12220-017-9763-x.

[7] H. Blum. "A Transformation for Extracting New Descriptors of Shape". In: *Models for the Perception of Speech and Visual Form.* Cambridge: MIT Press, 1967, pp. 362–380.

[8] J.-D. Boissonnat and M. Wintraecken. "The reach of subsets of manifolds". In: *J. Appl. Comput. Topol.* 7 (2023), pp. 619–641. DOI: 10.1007/s41468-023-00116-x.

[9] L. Calabi and W. E. Hartnett. "A Motzkin-Type Theorem for Closed Nonconvex Sets". In: *Proc. Amer. Math. Soc.* 19.6 (1968), pp. 1495–98. DOI: 10.2307/2036239.

[10] Y.-C. Chang et al. "Medial Axis Transform (MAT) of General 2D Shapes and 3D Polyhedra for Engineering Applications". In: *Geometric Modelling: Theoretical and Computational Basis towards Advanced CAD Applications. IFIP TC5/WG5.2 Sixth International Workshop on Geometric Modelling December 7–9, 1998, Tokyo, Japan.* Ed. by F. Kimura. Boston, MA: Springer US, 2001, pp. 37–52. DOI: 10.1007/978-0-387-35490-3_3.

[11] F. Chazal and A. Lieutier. "The λ-medial axis". In: *Graph. Models* 67 (2005), pp. 304–331. DOI: 10.1016/j.gmod.2005.01.002.

[12] S. W. Choi and H.-P. Seidel. "Hyperbolic Hausdorff Distance for Medial Axis Transform". In: *Graphical Models* 63.5 (2001),

pp. 369–384. ISSN: 1524-0703. DOI: 10.1006/gmod.2001.0556.
[13] P. Erdös. "Some remarks on the measurability of certain sets". In: *Bull. Amer. Math. Soc.* 51.10 (1945), pp. 728–731. DOI: 10.1090/S0002-9904-1945-08429-8.
[14] H. Federer. "Curvature measures". In: *Trans. Amer. Math. Soc.* 93 (1959), pp. 418–481. DOI: 10.2307/1993504.
[15] D. Fremlin. "Skeletons and central sets". In: *P. Lon. Math. Soc.* 74.3 (1997), pp. 701–720. DOI: 10.1112/S0024611597000233.
[16] P. J. Giblin and J. Warder. "Reconstruction from Medial Representations". In: *Amer. Math. Monthly* 118.8 (2011), pp. 712–725. DOI: 10.4169/amer.math.monthly.118.08.712.
[17] K. Houston and M. van Manen. "A Bose type formula for the internal medial axis of an embedded manifold". In: *Differential Geom. Appl.* 27.2 (2009), pp. 320–328. DOI: 10.1016/j.difgeo.2009.01.002.
[18] J.-C. Latombe. *Robot Motion Planning*. The Springer International Series in Engineering and Computer Science. New York: Springer US, 1991. DOI: 10.1007/978-1-4615-4022-9.
[19] T. Motzkin. "Sur quelques proprietes caracteristiques des ensembles convexes". In: *Att. R. Acad. Lincei* 21 (1935), pp. 562–567.
[20] H. Nass et al. "Medial Axis (Inverse) Transform in Complete 3-Dimensional Riemannian Manifolds". In: *2007 International Conference on Cyberworlds (CW'07)*. 2007, pp. 386–395. DOI: 10.1109/CW.2007.55.
[21] R. L. Ogniewicz. "Skeleton-space: a multiscale shape description combining region and boundary information". In: *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* (1994), pp. 746–751. DOI: 10.1109/CVPR.1994.323891.
[22] N. Papanelopoulos, Y. Avrithis, and S. Kollias. "Revisiting the medial axis for planar shape decomposition". In: *Comput. Vis. Image Underst.* 179 (2019), pp. 66–78. DOI: 10.1016/j.cviu.2018.10.007.
[23] R. T. Rockafellar and R. Wets. *Variational analysis*. Berlin Heidelberg: Springer, 2009. DOI: 10.1007/978-3-642-02431-3.
[24] W. van Toll et al. "The Medial Axis of a Multi-Layered Environment and Its Application as a Navigation Mesh". In: *ACM Trans. Spat. Algorithms Syst.* 4 (2018), pp. 1–34. DOI: 10.1145/3204456.
[25] F.-E. Wolter. "Cut loci in bordered and unbordered Riemannian manifolds". PhD thesis. Fechbereich Mathematik der Technischen Universitat Berlin, 1985.
[26] L. Younes. "The Medial Axis". In: *Shapes and Diffeomorphisms*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2019, pp. 57–72. DOI: 10.1007/978-3-662-58496-5_2.